

Team Charter

Quantum Routines: Implementation, Verification, and Resource Estimation

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Industry Mentor: Mariia Mykhailova (PsiQuantum)

Expertise and Growth

Kush Aggarwal — My research background is big data, Likelihood optimization and building inference frameworks, where I use clusters in CENPA as well as Lyon(France). I have a pretty good experience with python and linux as a whole. I have taken both 521/522 so remember most of the algorithms from there. I am excited about this opportunity with workbench, as we are building from the ground up to a full fledged optimization and resource calculations. Working with an inbuilt QDE is one of its kind and would help me learn how to work in parallel implementations and do multi -threading in quantum computing.

Thomas Huang — My research background is in quantum materials simulations, where I use high-performance computing due to the large computational cost of these calculations. I have basic experience with Python and HPC workflows. In this project, I hope to learn more about quantum algorithms, especially reversible computation and how algorithms are implemented in practice. I also want to gain experience working with GitHub and collaborative coding, which I have not done before.

Christopher Sharp — My expertise is in optimization e.g. through multi-objective cost functions, which could be applied to estimating and improving resources of quantum routines. I am familiar with Python and have some background in lower level classical computing. I think I could improve on both of those. I also haven't meaningfully contributed to a larger project yet, and while I am familiar with the concepts of CI/CD I haven't implemented them much in practice before. Finally, by understanding and implementing at least one quantum algorithm into Workbench, I hope to build a deeper understanding of reversible computing and quantum programming.

Support Needs

We could use help further understanding Workbench and how to implement algorithms in a continuous integration process.

Team Rules and Expectations

- How will we make sure everyone is heard?
 - A. Everyone should feel free to voice their opinion on a matter and we will try to solicit the viewpoints of team members who have been silent in case they hadn't felt comfortable speaking up.
 - B. We will also ensure each member has a chance to speak during meetings and encourage feedback on shared documents or code.
- How will we resolve team conflicts?
 - B. We will attempt to resolve conflicts through communication. If there remains an issue, we will discuss with our industry mentor and/or our instructor for mediation.
 - C. We will aim to reach a consensus and document decisions clearly to avoid repeated misunderstandings.

Communication Guidelines

- We will meet twice a week with our industry mentor and with the instructor. The former meeting will be remote. The latter meeting will be mixed remote and in-person as needed.
- We will use email and (as needed) GitHub Issues to communicate directly in between meetings.
- We will also use GitHub to track tasks and progress, and ensure important updates are shared clearly with all team members.

